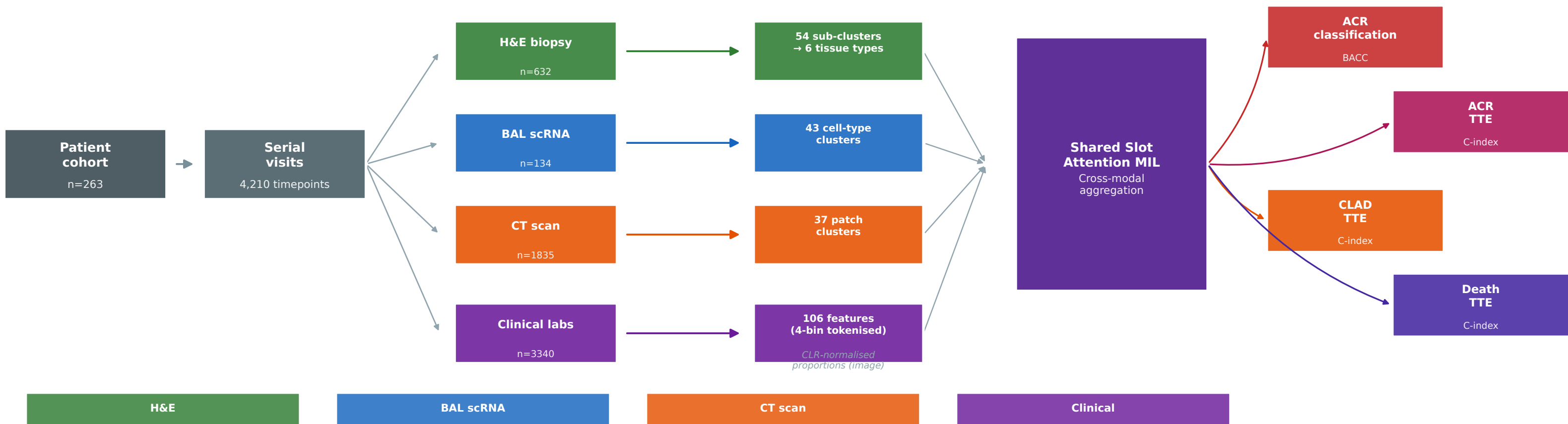
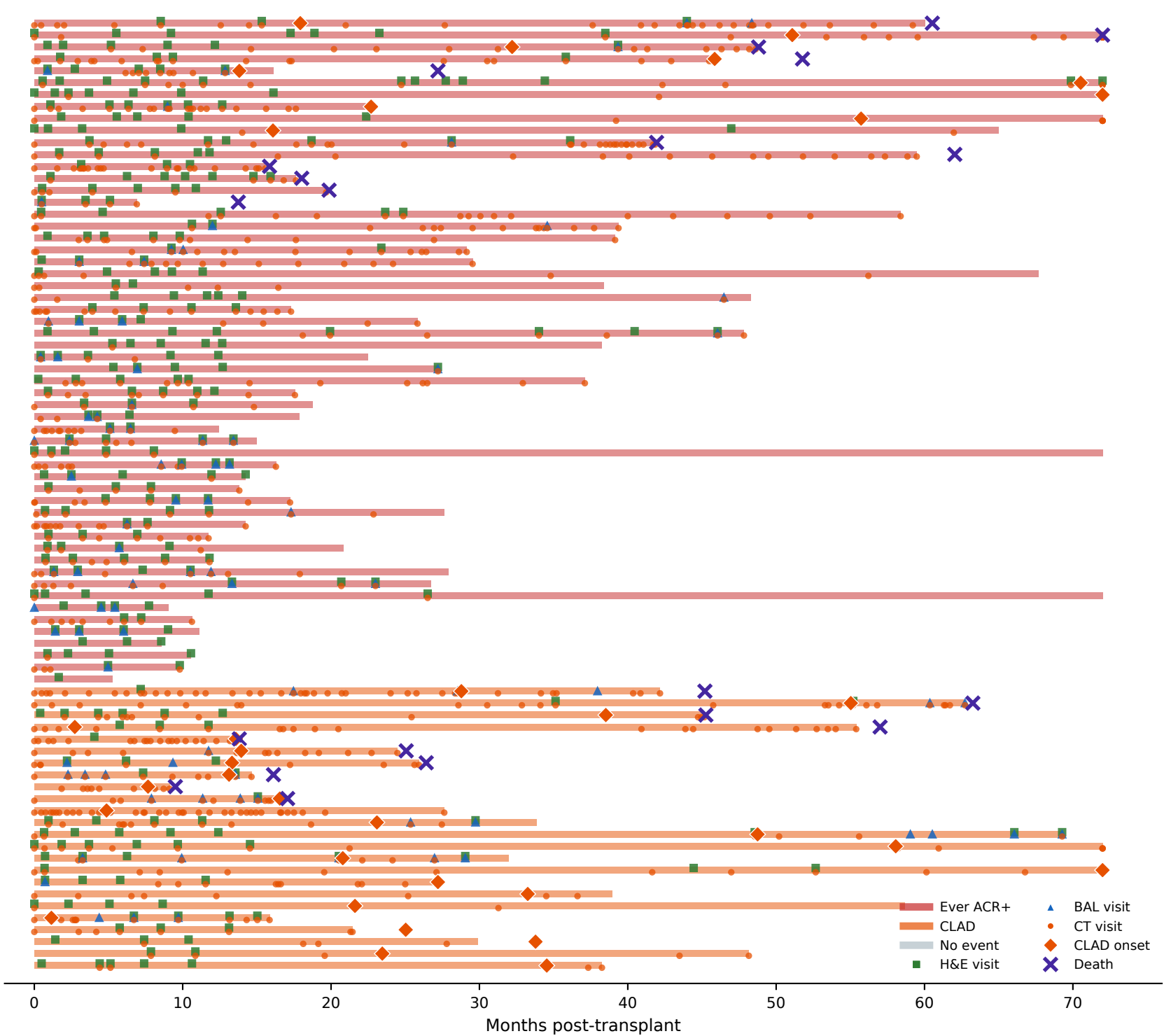


(A) Multimodal Multiple Instance Learning Framework — Lung Transplant Monitoring



(B) Patient follow-up timeline (top 80 patients, sorted by ACR/CLAD burden)



(C) Modality availability



(D) Cohort summary

Patients	263
Timepoints	4,210
ACR+ patients	56 (21%)
A0B0 timepoints	1,466
A1/A2 timepoints	167
CLAD events	37 (14%)
Death events	53 (20%)
H&E biopsies	632
BAL scRNA samples	134
CT scans	1,835
Clinical timepoints	3,340
H&E tissue types	6 (54 sub-cl.)
BAL cell types	43 clusters
CT clusters	37 clusters
Clinical features	106 (z-scored)
CV folds	4 outer (nested 5×4)

(E) ACR Classification

BACC: 0.74*

BAL Lymphocytes %

BAL Neutrophils %

Alveolar inflamed

Vascular H&E

FEV1

↓ protective

Decelerated FEV1 decline

Early innervation

Perivascular mononuclear infiltrate (A1/A2)

Endothelialitis — A2+ pathognomonic

↑ +ACR+

(F) ACR TTE

C-index: 0.71*

pseudoslope_FEV1

delta_FEV1

DSA count

CMV D+/R-

BAL CD4 T cell

↓ protective

FEV1 decline = BOS definition (≥20%)

PVC decline → RAS phenotype

Anti-DO DSA → sensitisation

CMV drives anti-HLA immunity

Effect of BAL CD4 on rejection pattern

↑ +event

(G) CLAD TTE

C-index: 0.73*

pseudoslope_FEV1

pseudoslope_FVC

BAL Neutrophils %

PGD T24 grade

CT air-trapping cl

↓ protective

Lung function → primary survival driver

Renal failure → mortality post-transplant

BAL >3% neutrophils → BOS predictor

PGD sets alloimmune milieu (OR ~3)

FEV1 trajectory

↑ +ACR+

(H) Death TTE

C-index: 0.69*

FEV1 (absolute)

GFR

donor_risk

age

pseudoslope_FEV1

↓ protective

Donor lung quality score

Recipient age → comorbidity burden

↑ +ACR+